PawWatch

Edge-AI Cat Health Monitor

GitHub Repo Demo Video

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Problem: Cats hide illness

Cats instinctively hide pain and illness; obvious symptoms such as vomiting may appear only when it is more severe [1], [2]. Earlier signs show up in daily habits: vets list changes in appetite, thirst and litter box use as warning signs [2], and each common disease changes them in its own way:

Disease	How common	Eating	Drinking	Litter
Kidney disease	Up to 30–40% of older cats may have it; only 3.6% are diagnosed [3], [4]		↑	↑
Hyperthyroidism	8.7% of cats over 10 [5], [6]	↑	↑	↑
Diabetes	About 1 in 200 cats [7]	↑	↑	↑
Urinary tract disease	A leading reason cats visit vets [8]			↑

A 2026 study of 136 cats found those with kidney disease used the litter box about twice as often, even in early stages [9]. A blocked urinary tract may kill within 24–48 hours; its first sign is many litter box trips [8].

Today’s smart devices each watch one spot

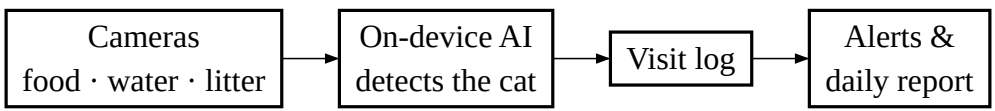
Each smart device on the market watches one spot, so watching food, water and litter takes several devices and apps.

	Food	Water	Litter
Smart feeders	✓	✗	✗
Smart water fountains	✗	✓	✗
Smart litter boxes	✗	✗	✓
Pet cameras (video only, no visit log)	✗	✗	✗
PawWatch	✓	✓	✓

However, illness shows up as a **combination**. For example, kidney disease raises both drinking and litter box use [3], so a water-only or litter-only device sees half the pattern. Once set up, PawWatch watches all three spots with one box, and the video never leaves home.

Solution: PawWatch

Owners are out all day, and no single device sees the whole pattern. **PawWatch** is a vision-based monitoring system: low-cost cameras watch the food bowl, water bowl and litter box, and an on-device AI accelerator recognizes the cat and logs every visit. Video never leaves home.



What it records

- **Visit log:** how often and how long the cat eats, drinks and uses the litter box
- **No-go zone alarm:** plays a sound when the cat jumps onto the counter

What it tells you

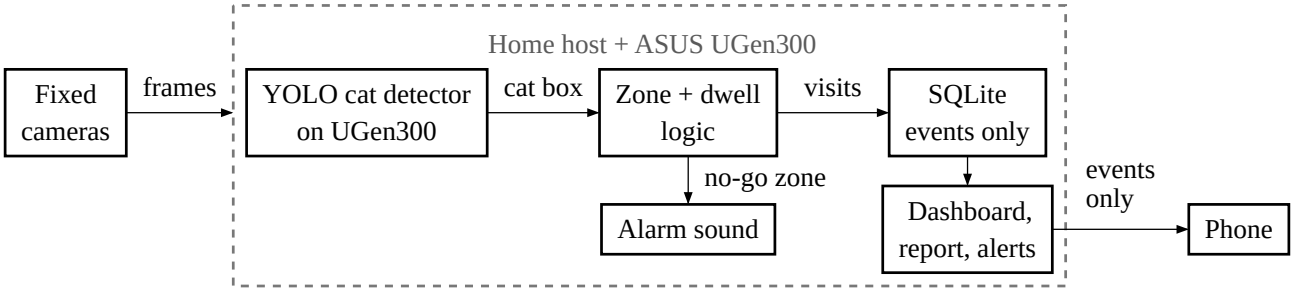
- **Anomaly alert:** ≥ 2 or ≤ 0.5 times the cat's 7-day average
- **Daily report:** a plain-language summary
- **Dashboard:** today's counts, timeline and 7-day trends

Why run AI at home, not in the cloud

	Cloud pet camera	PawWatch
Who sees your home video?	It's uploaded to the company's servers. At one home-camera maker, staff watched customers' private videos [10]	No one. Video never leaves home; only events like "14:32, drank for 25 s" are stored
Monthly fee	Smart features usually need a paid subscription	None. No servers to pay for
Internet goes down	Cloud features stop until it's back	Keeps logging; alerts are sent once it's back
Cat jumps on the counter?	Frames go to the server and back first	The alarm sounds right away, from the box at home

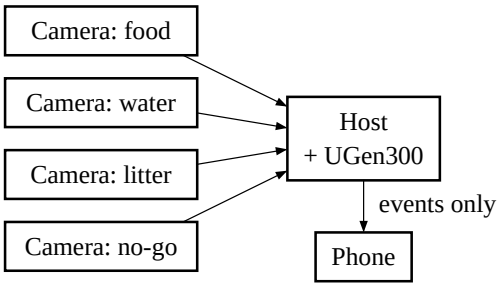
Running 24/7 is cheap: the AI accelerator draws 2.5 W, less than an LED bulb [11].

Architecture: video in, events out



- **Detect:** the host decodes each camera's video and the UGen300 runs a stock YOLO model that finds the cat. Frames live only in memory and are dropped after inference; nothing saves or uploads them.
- **Turn boxes into visits:** the cat's box is matched to the zones drawn for that camera (food, water, litter, no-go), and a visit is logged once the cat stays long enough. A no-go zone rings the alarm right away.
- **Store and report:** each visit is one row in a local SQLite database. The dashboard, anomaly alert and daily report read only these rows, so the only thing that can leave the home is an event like "07:30, ate for 31 s".

Hardware: one host, many fixed cameras



We mount **fixed cameras**, one at each spot we watch: the food bowl, the water bowl, the litter box and any no-go spot such as the kitchen counter. Each camera streams its video over the home network to **one small host computer** (a mini PC or Raspberry Pi).

The **ASUS UGen300** is plugged into the host over USB or M.2. The host decodes the video and hands frames to the UGen300, whose Hailo-10H chip (40 TOPS at 2.5 W [11]) runs the cat detector. The host then turns detections into visits, stores them and serves the dashboard. Only these events reach the owner's phone.

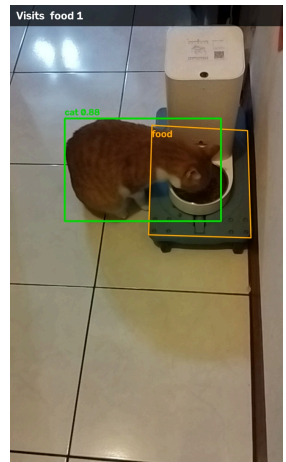
Why fixed cameras: zones are drawn in pixels, so a camera that never moves needs them drawn only once, at setup. One camera per spot also keeps the cat large in the frame, which makes detection reliable.

Detection: from pixels to “the cat is at the bowl”

A stock YOLO model, no training. We use YOLOv8n pretrained on the public COCO dataset, which already has a “cat” class. We keep only that class (score ≥ 0.4). No cat photos were collected and no model was trained or fine-tuned, so any home can use it out of the box.

Zones are drawn once. At setup, the owner clicks a polygon around each spot on a still frame (`draw_zones.py`); it is saved as one small JSON file per camera.

Which zone is the cat in? We take the highest-scoring cat box and measure how much of it overlaps each zone, as a share of the smaller of box and zone. The zone with the most overlap wins, if it reaches 20%. Measuring against the smaller shape lets a zone hug a small bowl, and a box that only grazes a zone’s edge doesn’t count.



Detection window: cat box, food zone and visit count

Event logic: from detections to visits

The event logic is pure code: it takes $(\text{video time}, \text{zone})$ pairs and returns visits, so it is tested without a model or video. Timestamps are **video time** (recording start + frame offset), so recorded footage replays exactly.



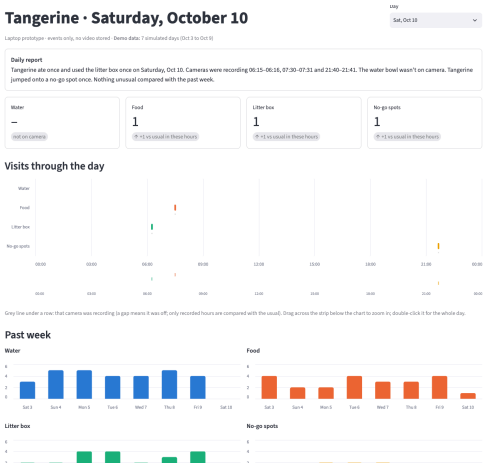
Minimum dwell	A visit counts after 5 s in one zone, so walking past a bowl isn't a meal. No-go zones need only 1 s: a quick jump still counts.
Gap tolerance	The cat may vanish for up to 5 s (turning, hidden by the litter box rim) without splitting one visit into two.
Live writes	A visit is stored as soon as it qualifies and its end moves forward while the cat stays, so the dashboard follows a live camera.
Alarm	Every frame in a no-go zone calls the alarm at once; a 30 s cooldown gives one sound per jump, not per frame.

Storage and outputs: what the owner sees

One table is the whole record. Each visit is one row in local SQLite: `camera, zone, start, end, duration, simulated`. A second table logs when each camera was recording, so hours with the camera off are never read as “didn’t drink”.

Anomaly alert. For each of food, water and litter, today’s count is compared with the cat’s own average over the previous 7 days, over the same hours, so a normal morning isn’t flagged for a missing dinner. An alert needs both ≥ 2 or ≤ 0.5 times the average and a difference of at least 3 visits, so 1 vs. 0 is not an alert.

Daily report and alarm. Template sentences turn the counts into plain language. Wording suggests watching the cat or seeing a vet, never a diagnosis. No-go jumps play a sound on the host and can send a text message (never a frame).

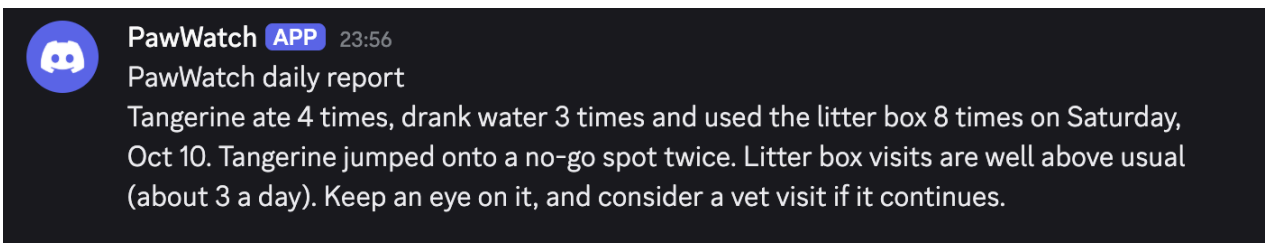


Dashboard: today’s counts, timeline, 7-day trend

Notifications: a text to your phone

Two moments, one message each.

- **No-go alarm:** the moment the alarm sounds, the host sends “your cat jumped onto a no-go spot at 14:32”. The 30 s cooldown keeps it to one text per jump.
- **Daily report:** each evening the host sends that day’s report, anomaly alerts included.



Daily report on Discord, with a litter box alert (simulated data)

Benefits for the cat: problems caught early, a safer home

Catch problems early

- **Early signs of many diseases**: appetite, thirst and litter use change before obvious symptoms
- **Emergencies in hours, not days**: an urgent change shows up in that evening's report, not after days of guessing

A safer, calmer home

- **No-go zones**: an alarm scares the cat off, and the owner gets a LINE or Discord text right away
- **Less stress**: the cat is observed at home, unhandled, so the record shows its normal behavior

> **Example: the litter box**
>
> Kidney disease doubles litter use even at early stages [9], yet most cases go undiagnosed [3], [4]. Found early, it is treatable: a kidney diet cut uremic crises from 26% to 0% in a 45-cat trial [12]. A blocked urethra also shows as many litter box trips, and can kill within 24–48 hours [8]

Benefits for the owner: rest assured, wherever you are

- **Peace of mind:** at work, on a business trip or in hospital, the cat is watched all day. Alerts reach the phone over LINE or Discord.
- **Facts instead of guesses:** 55.5% of owners did not recognize more thirst as a sign of dehydration [13]. PawWatch counts every visit for them.
- **One box, no subscription:** costs less than most products on the market. Pay once rather than subscribe monthly.
- **Privacy:** video never leaves home.

Earlier is cheaper

Caught early
A senior cat check-up: NT$3,000–8,000 [14]

Caught late
Emergency treatment can cost NT$30,000 or more at a 24-hour hospital [15]

Benefits for vets and beyond

Vets

- **A complete history**: a dated, detailed record can facilitate diagnosis
- **Recovery after treatment**: home data tracks recovery status

Resources

- **Low power**: all-day watching costs little energy
- **No internet demand**: everything runs locally (except LINE / Discord alerts and daily reports)

Market: 266k Taiwan homes with an older cat

Our goal: sell **5.3 k kits** in Taiwan in the first three years, **NT$79 M** in revenue.

Who	Homes	Market value
All households in Taiwan [16]	9.85 M	
… that keep a cat (10.4%) [17] TAM	1.02 M	NT$15.2 B
… whose cat is 7 or older (26%)* [18] SAM	266 k	NT$4.0 B
… that we reach in years 1–3 (2%) SOM	**5.3 k**	**NT$79 M**

Value = homes × NT$14,900 for one PawWatch Full Kit, bought once (see Business model, p. 16).

* 7 or older is when chronic disease becomes common; the survey covers dogs and cats together

Business model: buy once, no monthly fee

Choose one

- **PawWatch Full Kit, NT$14,900:** Raspberry Pi 5 + UGen300, preinstalled, with three night-vision cameras
- **PawWatch Minimal Pack, NT$7,900:** UGen300 + a USB stick that boots PawWatch on a spare PC; cameras not included

Add-on

- **Extra camera, NT$900:** additional night-vision camera to extend the coverage

Scalability

- **Extensible:** a new spot costs only a camera
- **Sustainable:** no subscription fee, no extra cost for long-term usage

Targeted audience roadmap

1. Cat owners · Year 1 → **2. Vet clinics** · Years 1–3 → **3. Device makers** · Year 3+

- **Who:** owners of cats aged 7+, 266k homes
- **Reach:** demo video shared in cat owner communities
- **Goal:** show in real homes that the alerts catch changes

- **Who:** about 1,800 animal hospitals [19]
- **Reach:** partner clinics recommend the kit for kidney disease, diabetes and after a urinary blockage
- **Goal:** a joint study with a vet school confirms the alerts

- **Who:** makers of pet cameras, feeders and litter boxes
- **Reach:** the vet-validated alerts from phase 2 make the licensing pitch
- **Goal:** PawWatch runs inside the devices owners already buy

References

[1] M. C. Evangelista et al., “Facial expressions of pain in cats: the development and validation of a Feline Grimace Scale,” *Scientific Reports*, 2019, [Online]. Available: https://doi.org/10.1038/s41598-019-55693-8

[2] VCA Animal Hospitals, “Recognizing Signs of Illness in Your Cat.” [Online]. Available: https://vcahospitals.com/resources/conditions-cat/recognizing-signs-of-illness-in-your-cat

[3] A. H. Sparkes et al., “ISFM Consensus Guidelines on the Diagnosis and Management of Feline Chronic Kidney Disease,” *Journal of Feline Medicine and Surgery*, 2016, [Online]. Available: https://journals.sagepub.com/doi/10.1177/1098612x16631234

[4] M. Conroy, D. C. Brodbelt, D. O'Neill, Y. Chang, and J. Elliott, “Chronic kidney disease in cats attending primary-care practice in the UK: a VetCompass study,” *Veterinary Record*, 2019, [Online]. Available: https://rvc-repository.worktribe.com/OutputFile/1382711

[5] M. J. Stephens et al., “Feline hyperthyroidism reported in primary-care veterinary practices in England,” *Veterinary Record*, 2014, [Online]. Available: https://rvc-repository.worktribe.com/output/1404762

[6] M. E. Peterson, C. A. Castellano, and M. Rishniw, “Evaluation of body weight, body condition, and muscle condition in cats with hyperthyroidism,” *Journal of Veterinary Internal Medicine*, vol. 30, no. 6, pp. 1780–1789, 2016, [Online]. Available: https://pmc.ncbi.nlm.nih.gov/articles/PMC5115195/

[7] D. G. O'Neill et al., “Epidemiology of diabetes mellitus among 193,435 cats attending primary-care veterinary practices in England,” *Journal of Veterinary Internal Medicine*, 2016, [Online]. Available: https://www.rvc.ac.uk/vetcompass/news/epidemiology-of-feline-diabetes-mellitus

[8] Cornell Feline Health Center, “Feline Lower Urinary Tract Disease.” [Online]. Available: https://vet.cornell.edu/departments-centers-and-institutes/cornell-feline-health-center/health-information/feline-health-topics/feline-lower-urinary-tract-disease

[9] Langenfield-McCoy et al., “Enhancing Detection of Chronic Kidney Disease Through Smart Litter Box Monitoring,” *Animals*, vol. 16, no. 9, p. 1319, 2026, [Online]. Available: https://www.mdpi.com/2076-2615/16/9/1319

References (2)

[10] U.S. Federal Trade Commission, “FTC Says Ring Employees Illegally Surveilled Customers, Failed to Stop Hackers from Taking Control of Users' Cameras.” [Online]. Available: https://www.ftc.gov/news-events/news/press-releases/2023/05/ftc-says-ring-employees-illegally-surveilled-customers-failed-stop-hackers-taking-control-users

[11] iThome, “ASUS UGen300 press release (NT$6,380, 2.5 W, 40 TOPS).” [Online]. Available: https://www.ithome.com.tw/pr/175676

[12] S. J. Ross et al., “Clinical evaluation of dietary modification for treatment of spontaneous chronic kidney disease in cats,” *Journal of the American Veterinary Medical Association*, vol. 229, no. 6, pp. 949–957, 2006, [Online]. Available: https://doi.org/10.2460/javma.229.6.949

[13] Sure Petcare, “Pet owners survey: how much water is your cat drinking? (12,776 owners; manufacturer survey).” [Online]. Available: https://www.surepetcare.com/en-us/advice-news/cat-care/general-cat-care/sure-petcare-pet-owners-survey-how-much-water-is-your-cat-drinking

[14] PetHealthMap, “Cat health check costs in Taiwan, 2026 (exam NT$300-800, blood tests NT$2,400-3,400, senior package NT$3,000-8,000).” [Online]. Available: https://pethealthmap.com/zh-TW/blog/cat-exam-cost-guide

[15] PTT WomenTalk, “Owner report: cat catheterization and tests NT$30,000 at a 24-hour animal hospital, NT$11,000 per day in hospital.” [Online]. Available: https://www.ptt.cc/bbs/WomenTalk/M.1693274189.A.D21.html

[16] M. of the I. Department of Household Registration, “Household registration statistics, December 2025 (9,851,532 households).” [Online]. Available: https://www.ris.gov.tw/info-liferay/app/channel/newsDetail/26007114

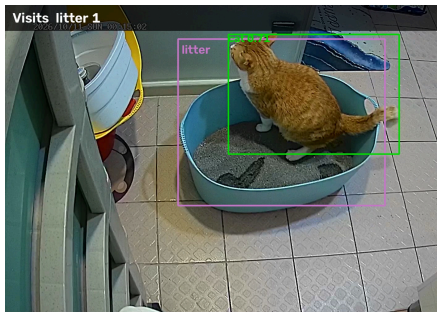
[17] Central News Agency, “Ministry of Agriculture 2023 dog and cat survey: 10.4% of households keep cats (10,348 phone interviews).” [Online]. Available: https://www.cna.com.tw/news/ahel/202405310157.aspx

[18] Business Weekly, “Over 26% of Taiwan's dogs and cats are 7 or older (TIER and FIRDI survey, 2024).” [Online]. Available: https://www.businessweekly.com.tw/focus/indep/6020425

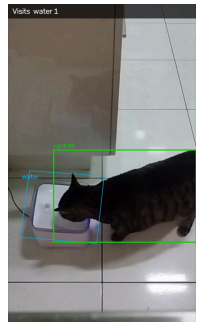
[19] Taipei Times, “Pet dogs and cats have outnumbered newborns (about 1,800 animal hospitals nationwide).” [Online]. Available: https://news.ltn.com.tw/news/focus/breakingnews/4641328

Appendix

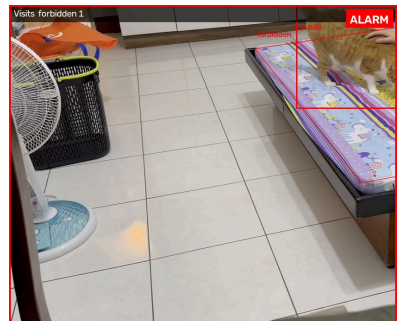
Detection screenshots



Cat in the litter zone



Cat at the water bowl



No-go zone: box turns red, alarm rings

Each window shows the cat box with its score, the camera’s zones and the visits counted so far. The windows are only displayed; frames are never written to disk.

Dashboard with an anomaly alert (simulated data)

This dashboard runs on **simulated** data only, to show what an alert looks like: a day where the litter box count reaches at least twice the cat's 7-day average.

The alert names the change and the usual count, and suggests keeping an eye on the cat or seeing a vet. It never names a disease.

